Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Fiscal year 8.1 · A law to be built

A balanced dice with six faces numbered from 1 to 6 is launched. X associated with the result
the rest of its Euclidean division by 3. For example, a result divisible by 3 gives $X = 0$.

1. For each of the six sides, indicate the value taken by X.

2. Presenting the distribution of probability of X in a table and verify that the sum of probabilities equals 1.

3. Calculate $P(X < 2)$ specifying which values of X are retained.

Exercise 8.2 · Two variance calculations

Random variable X takes the values -2, 0 and 4 with respective probabilities $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{1}{4}$.

1. Calculating His Expectation $\mu = E(X)$ and the moment $E(X^2)$.

2. Calculate $V(X)$ using the König-Huygens formula, then deduct the exact standard deviation.

3. Find variance from definition $V(X) = E((X-\mu)^2$Presentation of the three contributions the calculation.

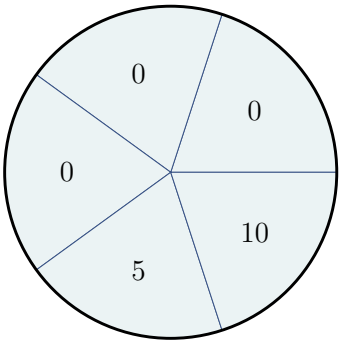
Exercise 8.3 · Fair play

A wheel has five sectors of the same angle, bearing numbers 0, 0, 0, 5 and 10.
the five equiprobable sectors. G, in monetary units.
To play, we pay a bet. m ; the net gain is therefore $X = G - m$.

1. To establish the law of G, then determine the bet m which makes the game fair, i.e. such that $E(X) = 0$.

2. After studying the variance, calculate the standard deviation of the gross gain. where the bet is fair.

3. Explain why changing the bet changes the expectation of the net gain but not its standard deviation.



The five sectors have the same angle. The three sectors bearing 0 are distinct and all must be Counted.

Exercise 8.4 · Same average, other risk

Two fictitious investments are proposed. The first one provides with certainty a gain $X = 1$. The second provides a gain $Y = -2$ with probability $\frac{1}{2}$ and one gain $Y = 4$ with probability $\frac{1}{2}$. The winnings are expressed in the same currency unit.

1. Compare the expectations of the two gains, then their variances.

2. One person states: "These investments are equivalent since they have the same average." Explain what this sentence omits and what the variance allows here to distinguish.

Exercise 8.5 · Linearity without independence

We are launching a balanced dice numbered from 1 to 6. X the result of the launch and it defines $Y = 7 - X$. The two variables are thus obtained from the same launch.

1. Calculate $E(X)$ and $E(Y)$, then $E(X + Y$ Check the linearity of expectation on this example.

2. Variables X and Y Are they independent? Justify using, for example, events $\{X = 1\}$ and $\{Y = 6\}$.

3. Does the linearity of expectation require independence? Conclude based on your results.

Exercise 8.6 · Minimum loss

A Random Variable X takes a finite number of values and has a expectation μ. For a real a, mean quadratic error associated with constant prediction is measured a by $E((X - a)^2)$.

1. Demonstrate that $E((X - a)^2) = V(X) + (a - \mu)^2$. We can write. $X - a = (X - \mu) + (\mu - a)$.

2. Deduct the value of a which minimizes this error, then the minimum value. Justify the uniqueness of the optimal choice.

Exercise 8.7 · Frequency of Letters

We want to measure the frequency of the letter "a" in a text. The denominator is the number of Alphabetical characters: spaces, numbers and punctuation signs are excluded.
are assimilated to the tiny ones; the accented letters remain distinct from "a".

1. Write a Python function that returns this frequency. Specify the selected result when the text contains no letters.
2. Test function with `Aba !`, then with `123 !` and with the empty string. Explain the results before running the programme.

Fiscal year 8.8 · Deferred review

There are two independent repeats of an experience whose probability of success is worth $\frac{1}{2}$. The random variable X counts the number of successes achieved.

1. Retrieving the law of X using the four possible results. Do not use a formula general distribution of binomial distribution.
2. Calculate $E(X)$ and $V(X)$ on the basis of this law.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, Exercise 28 p. 306, all questions. Calculate expectation from the table of law; maintain the probabilities associated with each value.

CQFD, exercise 31 p. 306, all questions; use the law of fiscal year 28. Calculate the then the standard deviation, respecting the rounding to the tenth requested by the manual.